Signals

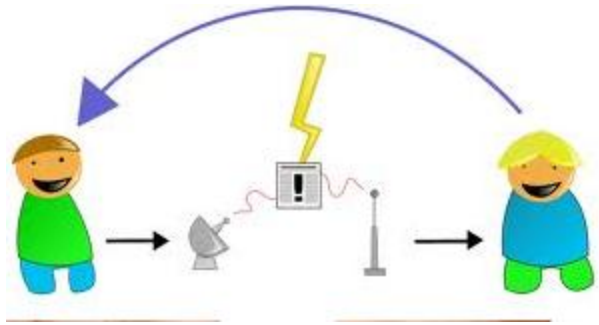
Analog Signals and Digital Signals, Characteristics of Analog Signals, Frequency Modulation (FM), Amplitude, Modulation (AM), Phase Modulation (PM), Quadrature Amplitude Modulation QAM

Application examples of signals









Definition

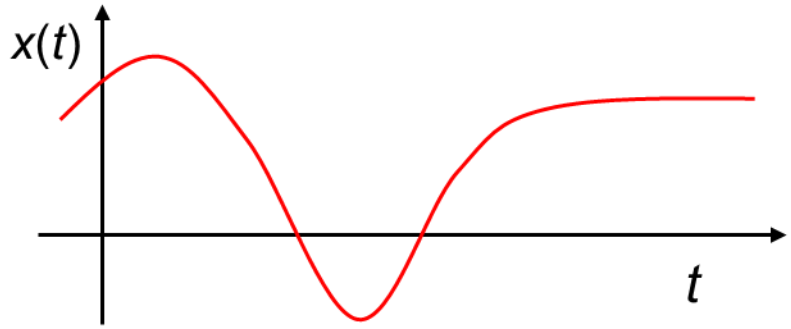
- Signal is a function of one (or) more independent variable which convey some information.
- Classification of Signals
  - Continuous time signals - x(t)

    (eg) Analog Signals
  - Discrete time signals - x(n)

    (eg) Digital Signals

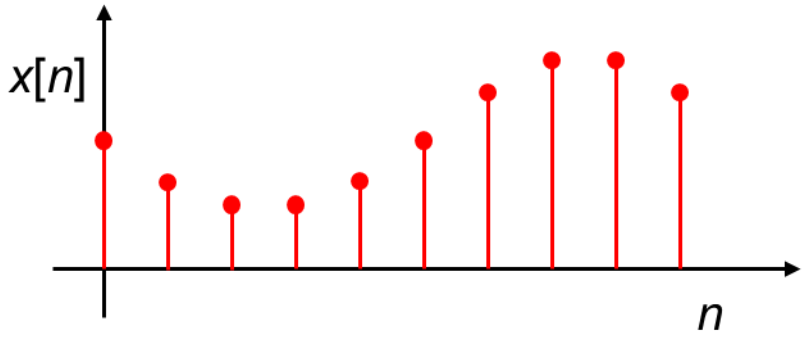
Continuous time signals

- Signal which are defined at every instants of time is known as continuous time signals.



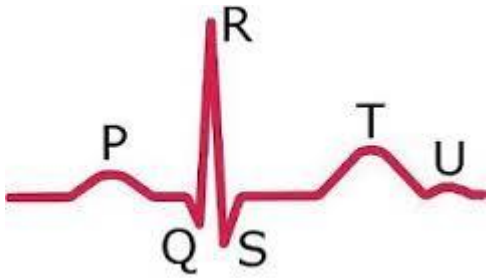
Discrete time signals

- Signal which are defined at discrete instants of time is known as Discrete time signals.



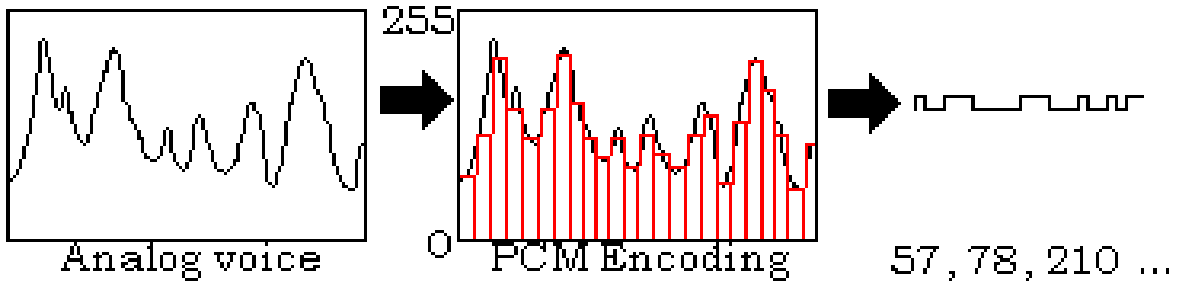
Examples of signals: ECG Signal

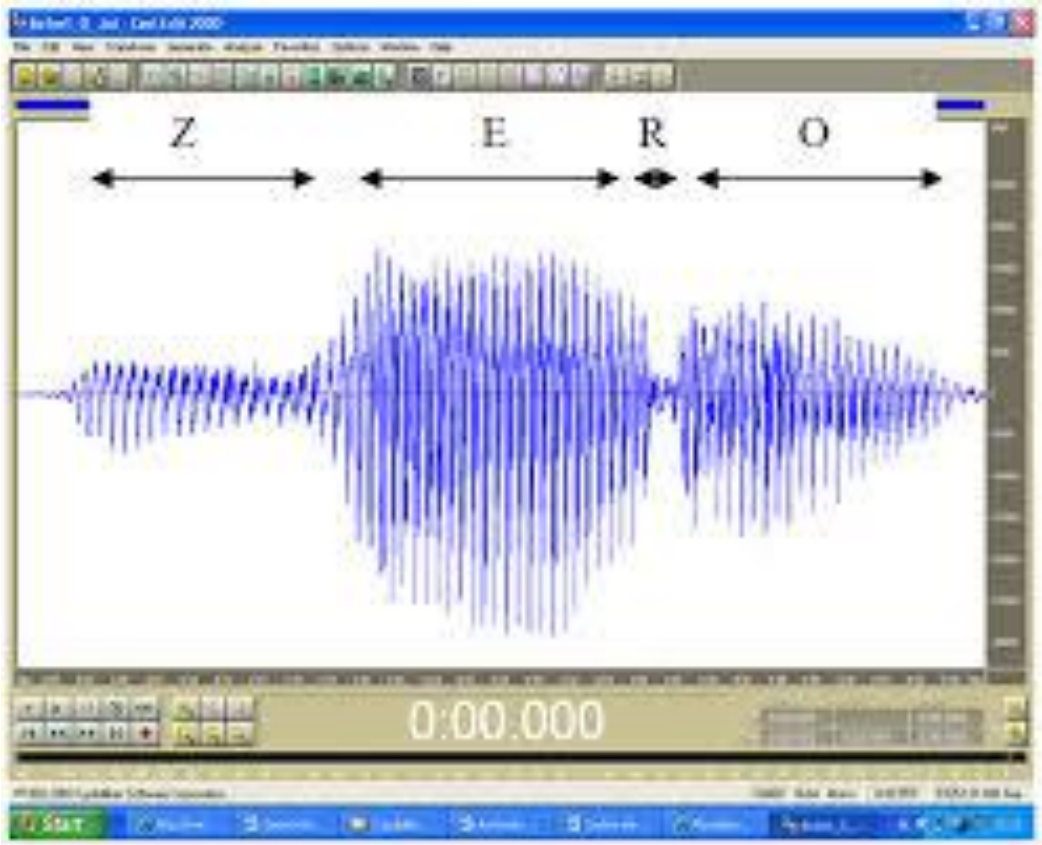




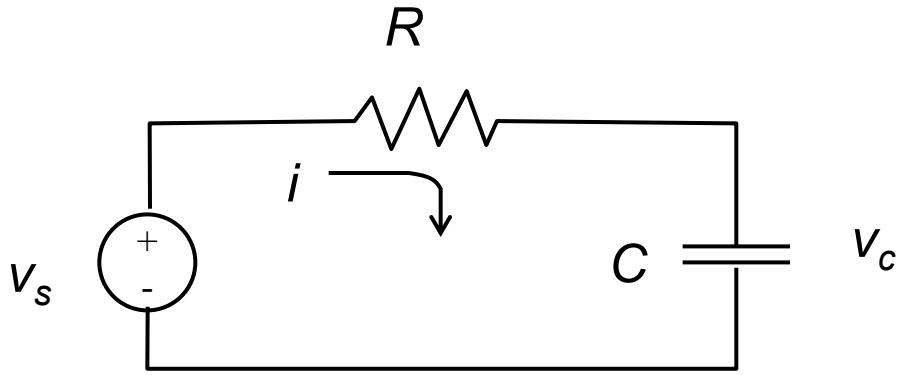


Voice to Digital conversion

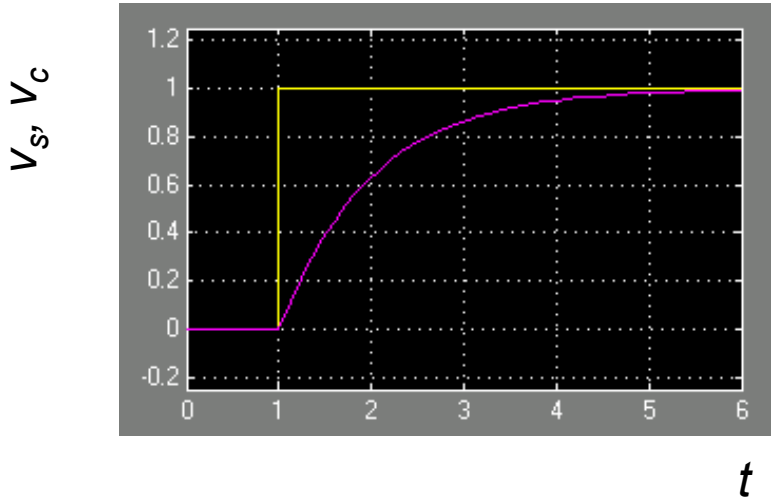



Z
E
R
O
0:00.000

Example: Signals in an Electrical Circuit

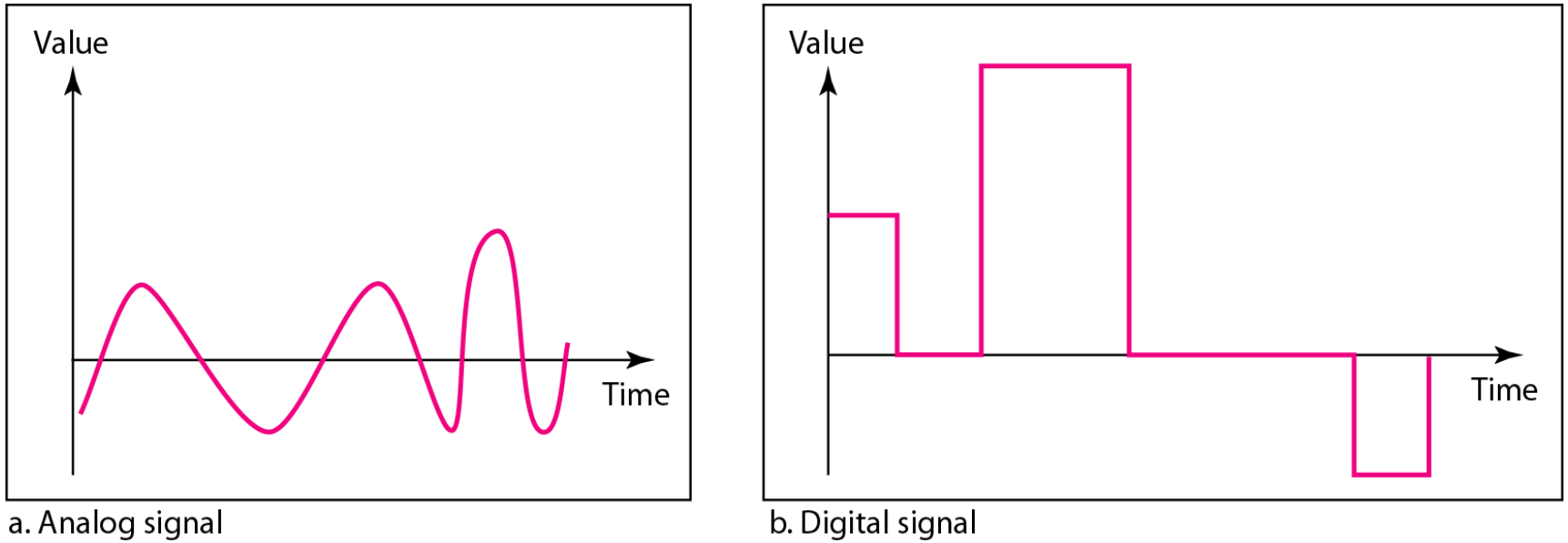


The signals v_c and v_s are patterns of variation over time



Signals can be analog or digital. Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values.

Figure *Comparison of analog and digital signals*



a. Analog signal

b. Digital signal

Figure *A sine wave*

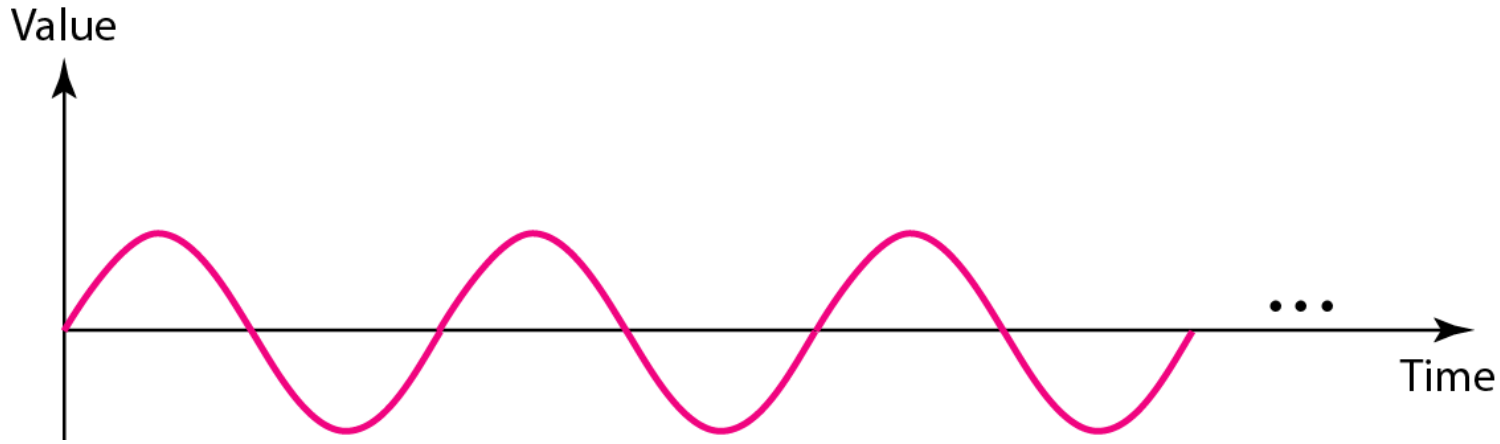
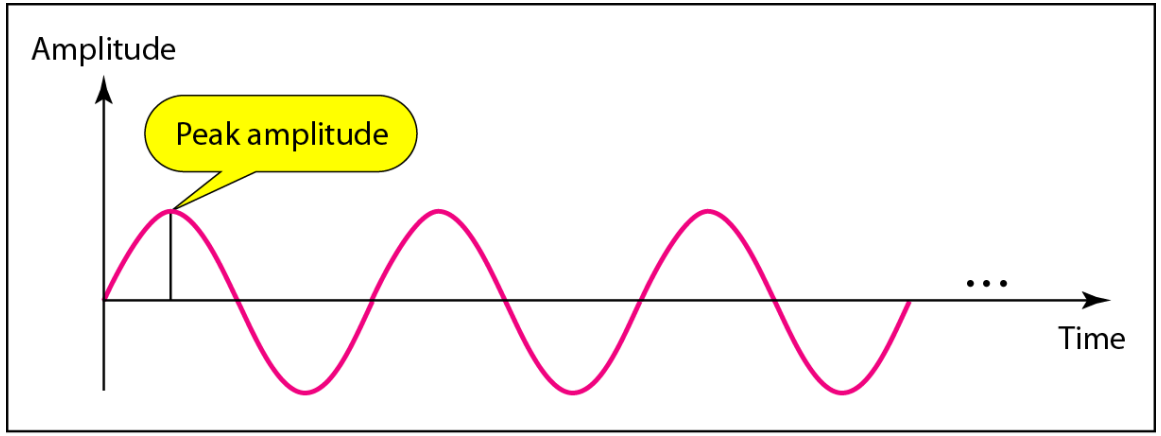
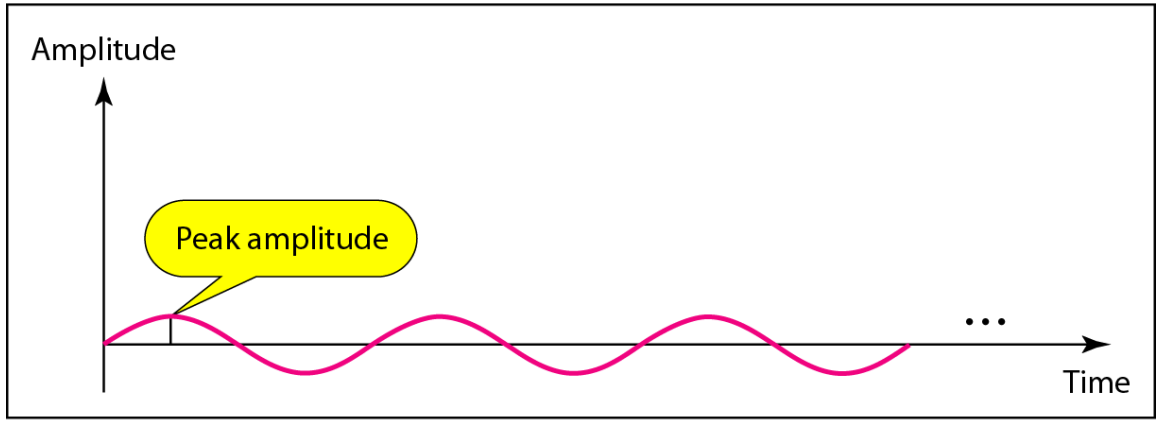


Figure *Two signals with the same phase and frequency, but different amplitudes*



a. A signal with high peak amplitude

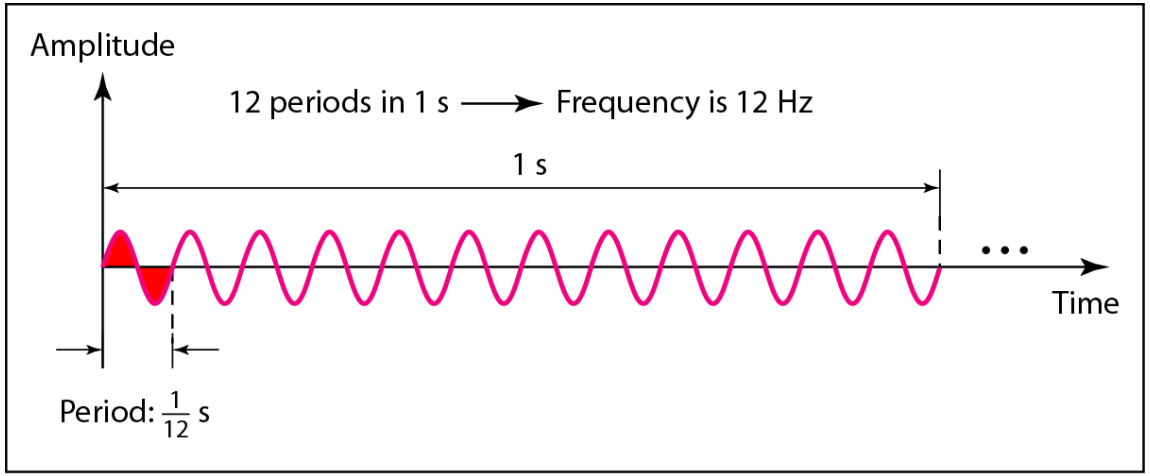


b. A signal with low peak amplitude

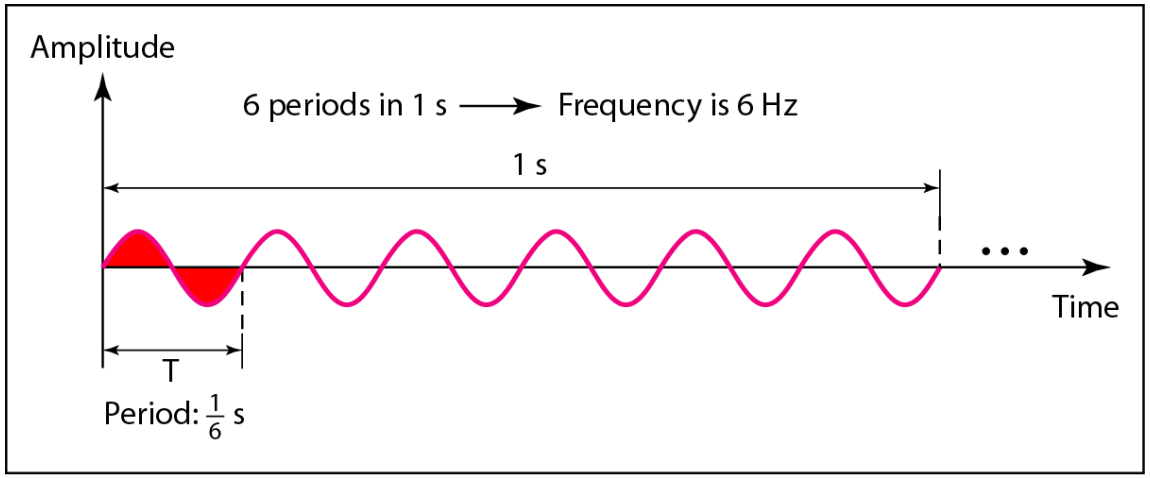
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Figure *Two signals with the same amplitude and phase, but different frequencies*



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Frequency is the rate of change with respect to time.

Change in a short span of time means high frequency.

Change over a long span of time means low frequency.

If a signal does not change at all, its frequency is zero.
If a signal changes instantaneously, its frequency is infinite.

Problems

- The power we use at home has a frequency of 60 Hz (50 Hz in Europe). Find the period of this sine wave

Contd…

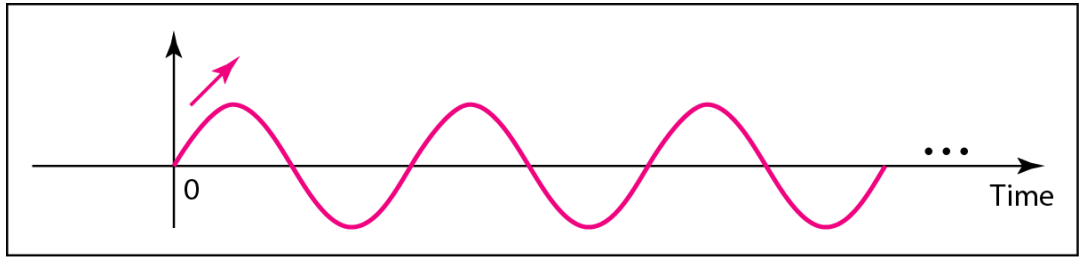
- Express a period of 100 ms in microseconds.

Contd…

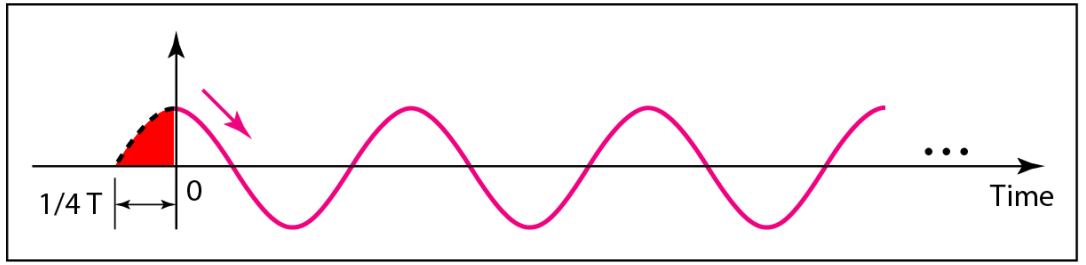
- The period of a signal is 100 ms. What is its frequency in kilohertz?

Phase describes the position of the waveform relative to time 0.

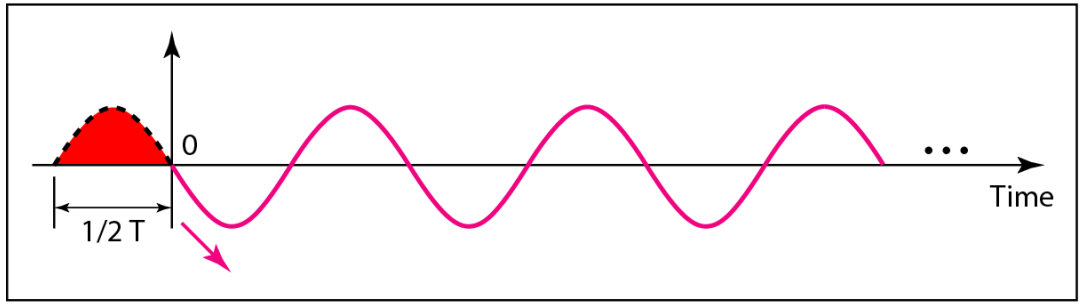
Figure *Three sine waves with the same amplitude and frequency, but different phases*



a. 0 degrees



b. 90 degrees



c. 180 degrees

Contd…

- a. A sine wave with a phase of 0° starts at time 0 with a zero amplitude. The amplitude is increasing.
- b. A sine wave with a phase of 90° starts at time 0 with a peak amplitude. The amplitude is decreasing.
- c. A sine wave with a phase of 180° starts at time 0 with a zero amplitude. The amplitude is decreasing

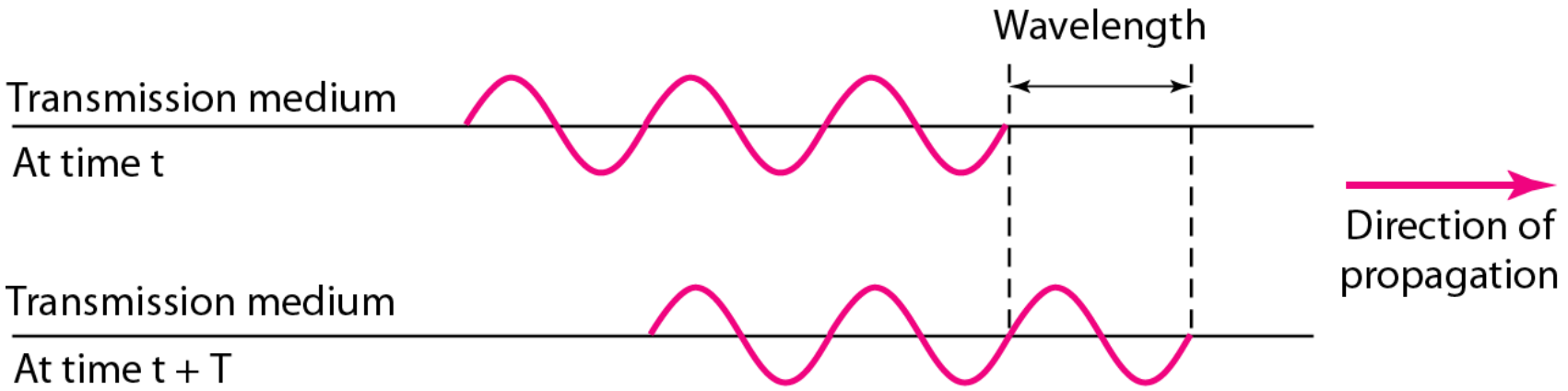
Contd…

- a. A sine wave with a phase of 0° is not shifted.
- b. A sine wave with a phase of 90° is shifted to the left by cycle. The signal does not really exist before time 0.
- c. A sine wave with a phase of 180° is shifted to the left by cycle. The signal does not really exist before time 0.

Problem

- A sine wave is offset cycle with respect to time 0. What is its phase in degrees and radians?

Figure *Wavelength and period*



The Taxonomy of Signals

- Signal: A function that conveys information

Time \ Amplitude	Continuous	Discrete
Continuous	analog signals	continuous-time signals
Discrete	discrete-time signals	digital signals